

Locating energy storage for offshore wind integration

A system-level analysis of offshore and onshore storage in the Dutch energy system

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Abstract

The Netherlands aims to achieve climate neutrality by 2050, with offshore wind energy forming a central pillar of its energy transition. Offshore wind capacity is projected to reach 21 GW by 2030 and 70 GW by 2050, supplying most of the country's electricity demand. However, the intermittent nature of wind generation and the limited capacity of onshore transmission networks increasingly lead to curtailment, grid congestion, and underutilization of offshore infrastructure. Energy storage can help mitigate these challenges by absorbing surplus electricity and supplying power during periods of deficit. Yet, the optimal siting of storage, offshore near generation or onshore near demand, remains unclear. Current studies provide contradictory results due to differing assumptions, insufficient spatial resolution, and limited applicability to the Dutch context. This research develops a spatially explicit, techno-economic energy system model using Calliope to evaluate the system-level conditions under which storage should be located offshore or onshore. The model explores both unconstrained optimization and policy-constrained scenarios to analyze how siting choices affect different KPIs. The outcomes are expected to reveal under what circumstances offshore storage becomes economically attractive compared to onshore alternatives and how these configurations impact the future Dutch electricity system dominated by offshore wind.

1. Introduction

1.1 Societal relevance and problem definition

Climate change, triggered by decades of accumulation of greenhouse gas emissions, presents an enormous challenge. The European Commission and the European Council launched the European Green Deal in December 2019 to limit global temperature rise to +1.5°C in reference to 1900 (European Council, 2024). The strategy sets a path for a green transition in the European Union (EU) to achieve climate neutrality by 2050 (European Commission, 2021).

Offshore wind energy is a key pillar of the future low-carbon European power system. The European Commission has set a target of installing 60 GW of offshore wind capacity by 2030 and 300 GW in 2050 (European Commission, 2020). The Netherlands alone plans to have more than 21 GW of offshore wind capacity in 2030 and 70 GW by 2050. To put this into perspective; the Dutch transmission system operator TenneT estimates that in 2030 offshore wind will generate roughly 75% of Dutch electricity demand, underscoring its scale and strategic importance for achieving climate neutrality (TenneT, 2025).

However, the large scale expansion of offshore wind also points to fundamental challenges in transforming energy systems. Wind is by nature intermittent: it produces unevenly with weather patterns and has times of excess generation and times of short supply. In surplus hours wind energy sometimes needs to be curtailed. In deficit hours on the other hand, dispatchable capacity, imports, or storage needs to meet demand in order to maintain reliability. These dynamics are amplified by the limited transmission infrastructure between offshore parks and the onshore grid: offshore export cables and substations are expensive and complex, while the onshore grid is already congested as well. As a result, assets risk underutilization during calm periods and congestion during peaks, undermining both the cost-effectiveness of infrastructure investments and the full societal value of renewable energy (TNO, 2022; ENTSO-E, 2024).

Energy storage is highlighted as a promising, maybe even unavoidable solution to address these issues. During periods of high wind power output, storage can absorb excess generation, and release energy during periods of shortage. Without sufficient storage, growing levels of curtailment, grid congestion, and revenue instability threaten both the economics of offshore wind and the reliability of the Dutch electricity system.

Integration of energy storage directly coupled to offshore wind farms is receiving increased attention in literature. While offshore storage is still in a developing phase, recent studies highlight its potential to fundamentally reshape offshore wind integration. Through reduction of curtailment at the source, revenue smoothing, and reducing pressure on onshore grids, offshore storage could reposition offshore wind farms as more self-reliant energy hubs rather than simple generation sites (Li & DeCarolus, 2015; Wiegner et al., 2024). On the other hand, offshore storage faces higher installation costs, marine operational risks, and regulatory uncertainty. Onshore storage, by contrast, is already commercially viable, easier to install, and critical for balancing demand and supply in existing grids. The decision of where to locate storage is therefore not a minor technical detail but a strategic choice with far-reaching implications for the cost, resilience, and spatial footprint of the Dutch energy system.

Despite the importance of this challenge, academic literature is fragmented. Some case studies find no business case for offshore storage (Atherton et al., 2023; Van den Haak, 2023);, while others state it is economically viable (Bijl, 2019; Juhlin et al., 2024). The difference in results is largely driven by

methodological choices, assumptions and site specific characteristics. In one of the most comprehensive studies thus far, Wiegner (2025) uses a system-level model with 2000 energy systems variations to conceptually give insight in optimal storage siting. However, the model lacks spatial detail, important offshore characteristics and the high shares of offshore winds that are expected in the Netherlands. This limits its ability to guide the actual onshore/offshore trade-off in the Netherlands. On the other hand, infrastructure-detailed models, like the work of Bos (2025), capture high-resolution grid dynamics but do not directly address the offshore versus onshore storage trade-off.

This thesis intends to fill these gaps with an evaluation of the techno-economic system-level impacts of onshore and offshore storage in Dutch offshore wind integration. Through coupling a spatially detailed, multi-node modelling framework with realistic grid constraints, curtailment dynamics, and diverse storage technologies, the research will try to provide insights on how offshore wind can be integrated together with storage.

Overall, the urgency of climate change, the Dutch ambitious offshore wind targets, the problem of intermittency, and the transmission constraints all together emphasize the social importance of this study. Making sure that offshore wind is integrated properly is not only vital for meeting climate goals but also for preserving the affordability, reliability, and sustainability in the Dutch energy transition.

1.2 Relevance to master program

The aim of the thesis is to examine the techno-economic trade-offs of offshore vs. onshore co-location of energy storage in the integration of large-scale offshore wind into the Dutch energy system. The thesis tries to provide design-oriented insights into how the siting of storage impacts overall system cost, reliability, and spatial feasibility.

The topic can be described as a typical problem within the Complex Systems Engineering and Management (CoSEM) programme. Offshore energy integration is a classic socio-technical challenge: large-scale technical infrastructure, many stakeholders, market and regulatory frameworks, and long-term social objectives such as decarbonization and energy security. The research requires quantitative technical system modelling and qualitative attention to institutional and economic contexts.

The thesis therefore fits the Final Attainment Levels of CoSEM. It integrates system design and techno-economic analysis with organizational and policy considerations, and it offers actionable recommendations for future energy system designs. The research is therefore not just an academic exercise, but also a design contribution to the socio-technical problem of integrating offshore wind in a sustainable and resilient way.

The remainder of this proposal is structured as follows. Chapter 2 reviews relevant literature and identifies research gaps. Chapter 3 describes the research design, model framework, and sub-questions. Lastly, chapter 4 presents the planning and timeline. Together, these sections outline the approach for analysing optimal energy storage siting.

2. Literature review

2.1 Content table

Nineteen relevant papers are selected using the strategy explained in the Appendix. Table 1 shows a content overview per paper.

No	Author(s)	Year	Title
1	Angizeh et al.	2023	Multi-node UC model (US case): distributed vs centralized storage siting
2	Atherton et al.	2023	Onshore hybrid parks (Sweden/UK): BESS vs grid expansion economics
3	Baldi et al.	2022	Offshore wind with batteries, H ₂ , and ammonia: timescale complementarity
4	Bijl	2019	MSc thesis: Offshore vs onshore energy storage at a single Dutch offshore wind farm
5	Bos	2025	MSc thesis: High-resolution Calliope model – BESS siting for Dutch grid congestion
6	Bulder et al.	2021	Dutch policy roadmap: role of offshore storage in future value creation
7	Juhlin et al.	2024	Increasing revenues of offshore wind farms with co-located energy storage systems
8	Kazemi-Robati et al.	2025	Hybrid BESS–H ₂ systems: offshore vs onshore vs split siting (technical MILP)
9	McKenna et al.	2021	Balmorel model: repurposing Danish O&G assets for offshore hydrogen
10	NSWPH consortium	2022	Industry discussion paper: Offshore grid-integrated Power-to-Gas
11	Qazi et al.	2025	Offshore BESS in NL/BE hybrid parks: congestion and ancillary services
12	Rabanal et al.	2024	Review: Storage for offshore wind services (BESS, H ₂ , CAES) – offshore vs onshore siting (technical)
13	Singlitico et al.	2021	Techno-economic analysis of offshore vs onshore electrolysis hubs
14	Travaglini et al.	2025	Turbine-level Dutch offshore study: offshore hydrogen vs onshore HVDC
15	Varotto et al.	2025	Offshore hybrid parks (wind + wave): electricity vs hydrogen transport routes
16	Van den Haak	2023	MSc thesis: Port-based onshore electrolysis vs offshore hydrogen hubs in NL
17	Van Stralen et al.	2025	OPERA study: Offshore hydrogen dominance by 2040, role of salt cavern storage
18	Wang et al.	2019	Review: Marine renewable energy storage technologies (batteries, CAES, PHS, H ₂ , buoyancy)
19	Wiegner	2025	PhD thesis: AdOpT-NET0 system model – offshore wind integration and role of storage

Table 1 – Overview of selected papers

The topics on which an article is selected form the basis of the literature review as they indicate the core concepts. Table 2 indicates which core concepts are addressed in which paper. Spatial resolution refers to the level of geographical detail in a model; higher resolution means more locations and a more detailed representation of grid and resource variations. System-level models, in contrast to isolated case studies or ‘island models’, include the full energy system and its interactions.

Core concept	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Dutch/North Sea				x	x	x			x	x			x		x	x		x
Offshore vs onshore				x			x				x	x	x	x	x			x
Hydrogen	x		x		x		x	x	x		x	x	x	x	x	x	x	x
Other storage tech.	x	x	x	x	x	x	x			x	x			x			x	x
High spatial resolution	x				x								x					
System-level modeling	x							x								x		x

Table 2 – Literature review content table

2.2 Literature synthesis

The rapid expansion of offshore wind in the North Sea raises fundamental questions about where to best place the storage capacity needed to integrate such large volumes of variable generation. While storage has traditionally been considered an onshore asset, the scale and distance of future offshore buildout, combined with growing grid bottlenecks at landing points, have brought the possibility of offshore storage into sharper focus. The debate is relatively recent, and the academic literature is still fragmented, with different streams of work emphasizing different aspects of the offshore vs. onshore trade-off.

2.2.1 System-level and spatially detailed modelling approaches

Large system-level analyses, such as those by Wiegner (2025), demonstrate the importance of storage to achieve deep decarbonization and a reliable North Sea energy system. Wiegner’s model describes the value of short-, medium-, and long-term storage technologies for balancing renewables with a comprehensive sensitivity analysis of 2,000 energy system configurations. This approach provides interesting conceptual insights about the systemic conditions under which offshore or onshore storage is preferred. However, the model is high-level on a few important aspects. It represents just one offshore and one onshore node, makes general distance assumptions (50–300km), and does not use the high shares of offshore winds expected in the Netherlands. As a result, outcomes are strongly driven by cost assumptions and backup availability (Wiegner, 2025) and cannot represent the locational, infrastructural, and regulatory problems that determine the actual onshore/offshore trade-off in the Netherlands.

At the other end of the spectrum, detailed spatial models that account for infrastructure such as Bos (2025), highlight how storage can relieve congestion when sited at provincial nodes near offshore wind landing zones. These works show that storage siting matters, and that transmission expansion alone cannot eliminate all bottlenecks. However, they remain limited to onshore batteries, and therefore overlook the potential role of offshore storage, as well as the wider range of storage technologies that could contribute to system flexibility at different timescales. Together, these two directions in literature illustrate that while offshore storage is widely acknowledged as potentially relevant, its implications have rarely been tested within system-level, spatially resolved models.

2.2.2 Mixed evidence on storage siting across different contexts

Outcomes from single case and broader techno-economic studies point in multiple directions regarding the siting of storage. Bijl (2019) examined a case of a Dutch offshore wind farm and compared onshore and offshore storage. His results suggest that offshore co-location of batteries reduces curtailment and often improves NPV, although the analysis might underestimate the additional capital costs of offshore infrastructure. Qazi et al. (2025) find similar results for hybrid North Sea parks combining wind and solar energy. They show that offshore batteries could indeed be profitable, with an important reason being the reduction of local congestion, while also providing stabilizing services such as frequency response. Analyses of Portuguese offshore parks that combine wind and wave power with multiple forms of storage, including batteries and hydrogen, also find positive results for offshore storage. They state batteries can smooth power output and that offshore hydrogen transport via pipelines can be competitive with electricity cables, although profitability highly depends on cost assumptions for shipping frequency and pipeline expansion (Varotto et al., 2025). Travaglini et al. (2025) also find that centralized offshore hydrogen production can reach lower costs than onshore production, mainly due to savings in offshore electricity infrastructure, but their scope is limited to hydrogen. Juhlin et al. (2024) expand this finding by modeling a German offshore wind farm with subsea hydro-pneumatic storage technology (FLASC) and comparing this to an onshore battery. They find that both are preferable compared to no storage. Moreover, subsea hydro-pneumatic storage is found to economically outperform an onshore battery, but this is due to its longer lifetime and no degradation rather than the siting choice.

While the previous studies highlight potential benefits of offshore storage, others studies emphasize significant risks and constraints. Some studies underscore that offshore options require higher upfront investment and are subject to technical risks, which may offset their benefits under less favorable conditions. In an evaluation of electrolysis in Dutch ports and offshore hubs, Van den Haak (2023) finds that onshore locations are still cheaper when land is available and grid access is easy. He argues offshore hubs are mainly useful in cases of serious space shortages. Similarly, Atherton et al. (2023) examine forty-seven Swedish wind farms and show co-location with onshore batteries to perform better than expensive grid reinforcement, but find no suggestion that shifting storage offshore would lead to added benefit there. These non-uniform outcomes demonstrate that neither onshore nor offshore storage is intrinsically superior. Their relative value depends significantly on assumptions made, local grid condition, and the balance between infrastructure costs and avoided constraints.

2.2.3 Hydrogen and importance of storage siting

Hydrogen has received particular attention in recent literature, especially as a mean for long-term storage. Studies such as Singlitico et al. (2021), van Stralen et al. (2025), and McKenna et al. (2021) show that offshore electrolysis may become preferable once transmission costs and spatial limitations are considered. Still, also for hydrogen there is no consistent easy answer. Moreover, these studies are often restricted to hydrogen alone and do not consider shorter-term storage technologies such as batteries or compressed air. They also tend to work with simplified time representations instead of modelling hourly system dynamics. As a result, they provide only a partial picture of the broader system integration challenge.

Other recent studies further indicate that location and technology choices matter. Kazemi-Robati et al. (2025) demonstrate that different siting strategies for hybrid battery–hydrogen storage configurations (offshore, onshore, or split) can change technical outcomes such as curtailment reduction and power

smoothing. This argument is reinforced by Angizeh et al. (2023), who demonstrate that distributing storage across multiple grid nodes yields greater benefits than centralizing it in a single location. Baldi et al. (2022) stress that different storage technologies have different roles, batteries are best suited for short time balancing and hydrogen and ammonia perform better at longer horizons. Together, these works highlight the importance of siting and technology mix, but they either lack techno-economic dispatch optimization, leave out offshore placement as a real option, or focus on small rather than system-level contexts.

2.2.4 Industry perspectives and remaining knowledge gaps

The mixed findings in academic literature are also found in reviews and industry perspectives, which further underline the uncertainty in this space. Wang et al. (2019) and Rabanal et al. (2024) provide broad overviews of marine storage options but stop short of evaluating their system-level roles. The NSWPH consortium (2022) and Bulder et al. (2021) suggest offshore storage as a likely driver of future system value, but these assessments are qualitative. Dutch practitioner perspectives, such as those discussed in a TKI webinar (2025), add to this uncertainty by warning that offshore batteries may face high curtailment risks if not carefully integrated into the system. Taken together, the literature signals that offshore storage could play a role in the Dutch energy transition, yet the conditions under which it is preferable to onshore siting remain ambiguous.

This ambiguity defines the knowledge gap. Offshore storage has been shown to yield benefits under specific assumptions, while other work stresses its cost and complexity. What is missing is a systematic, spatially resolved, and technology-diverse assessment of storage siting that accounts for offshore wind buildout, grid bottlenecks, and the complementary roles of short-, medium-, and long-duration storage. This thesis aims to provide such an analysis, moving beyond isolated case studies and system models that lack spatial resolution, to test whether offshore storage actually contributes to a cost-effective and reliable Dutch electricity system. The research aims to answer the following research question:

‘What conditions govern the cost-effective siting of energy storage offshore versus onshore, and what are the system-level implications of this siting choice in the context of increasing offshore wind capacity in the Dutch energy system?’

In this proposal, cost-effectiveness refers to minimising total system cost, including investment, operation, and transmission expenditures, while maintaining system reliability. Reliability entails the ability of the power system to meet demand in all hours without unserved energy.

3. Research methodology

In the last chapter, the following research question was formulated:

‘What conditions govern the cost-effective siting of energy storage offshore versus onshore, and what are the system-level implications of this siting choice in the context of increasing offshore wind capacity in the Dutch energy system?’

This thesis adopts a modelling approach to outline and analyse the conditions under which energy storage should be located offshore or onshore within the Dutch power system in 2040, as offshore wind rapidly grows. The approach is explanatory and quantitative with two complementary experiments: (i) condition-discovery runs where siting and sizing are endogenous (unconstrained optimum), and (ii) policy-constrained configuration runs (fixed shares or siting rules) that exhibit trade-offs under recognizable system conditions. The experiments are grounded in a targeted synthesis of the literature (for definition of mechanism, drivers, and system boundaries). An existing spatially explicit energy-system model in Calliope (Bos, 2025; Hasselaar, 2025) is extended with offshore nodes, landing points, and energy storage technologies that are currently not represented.

A spatially explicit optimization-based energy-system model is suitable for the question since onshore/offshore siting depends on system-level interactions among offshore hubs, landing points, onshore grid, and storage, all under least-cost capacity expansion and dispatch. Calliope facilitates an open, reproducible setup with explicit technology definition, constraints, and hourly multi-node resolution (Pfenninger & Pickering, 2018). Even though a system-level model as this is always subject to simplifications, it will provide more complete insights than a smaller more detailed model without system interactions.

3.1 Sub-questions

The main research question is addressed via five sub-questions:

- **SQ1.** *Which mechanisms and factors drive the offshore-versus-onshore storage siting trade-off?*
- **SQ2.** *How can a spatially explicit energy-system model capture these mechanisms and represent the differences between onshore and offshore storage?*
- **SQ3.** *Under what combinations of the identified drivers (SQ1) does the unconstrained, cost-minimizing model endogenously allocate storage offshore rather than onshore, and to what extent?*
- **SQ4.** *When offshore/onshore storage shares (or siting rules) are fixed exogenously, how do power-system outcomes change across representative system contexts?*
- **SQ5.** *What (quantified) insights regarding the onshore/offshore trade-off do these results provide for designing a cost-effective and reliable Dutch electricity system with large-scale offshore wind and storage integration?*

3.1.1 SQ1 – Mechanisms & scope

Sub-question 1 identifies mechanisms/factors (e.g. techno-economic, spatial, policy, environmental) behind the onshore/offshore siting trade-off that need to be represented by the model. This is done through targeted desk research using academic and grey literature and datasets from, for example, IEA and IRENA technology cost reports, ENTSO-E and TenneT for grid topology and landing-point data, ENTSO-E Transparency or Netbeheer Nederland for electricity demand profiles, and PBL or RVO for policy and spatial-planning information (these are indicative, not limited to). A lot of energy system characteristics can, after careful validation, also be taken from the existing Calliope model (Bos, 2025; Hasselaar, 2025). Optionally, experts within the OESTER consortium can validate uncertain mechanisms (TNO, 2025).

A conceptual cause–effect diagram will summarize how these mechanisms/factors influence the siting decision. In addition, the mechanisms/factors will be ranked on their importance, based on (i) expected quantitative influence and (ii) data availability. The data (i.e. parameter ranges) will also be mapped through targeted desk research. In parallel, KPI's on which to value the siting choice will be selected (e.g. total system cost, reliability, CO₂ emissions). Again, this will be done through targeted desk research and literature review. Ultimately, SQ1 will provide mechanisms/factors and KPI's that affect the offshore/onshore trade-off (feeding SQ3 & SQ4) and systemic context variables that affect all storage (feeding SQ4).

3.1.2 SQ2 - Model representation & validation

In sub-question 2, the most important mechanisms of SQ1 are implemented in a model that makes a particular distinction between offshore and onshore storage. First, a table with model boundaries and necessary assumptions/simplifications for the mechanisms to be implemented in Calliope will be constructed. Next, the existing Calliope multi-node representation of the Netherlands (Bos 2025; Hasselaar 2025) is extended with offshore nodes, landing points, and cables, while storage technologies are available at both offshore and onshore nodes with distinct techno-economic characteristics. Inputs required follow from SQ1 (but could for instance include the network topology and transfer capacity between offshore nodes and landing points, technology cost/efficiency ranges, hourly demand and RES profiles, policy or market-design relevant parameters, etc). The final result will be a baseline model that will be validated to provide logical responses and compared with historical dispatch data.

3.1.3 SQ3 - Condition discovery

Sub-question 3 aims to discover when, and by how much, the cost-minimizing model prefers offshore rather than onshore storage. This is done through a parametric experiment with endogenous storage siting and sizing, testing the combination of three to four factors from SQ1 (e.g. offshore CAPEX premium, landing-point capacity). To maintain reasonable computational power, a 7×7 grid or Latin Hypercube Sampling of the selected input parameters will be used (McKay et al., 1979; Montgomery, 2013), utilizing parameter ranges and central values from SQ1. Outputs can for instance be visualized in heat maps; for each combination of key conditions the color shows how much storage the model would place offshore. Alongside the map, other KPI's are reported (KPI's will follow from SQ1, but could for instance be curtailment, transmission build/avoidance, storage MW/MWh by node).

3.1.4 SQ4 - Policy-constrained configurations

Where SQ3 investigated a theoretical, unconstrained optimum, SQ4 maps the trade-offs that arise with fixed shares of offshore storage. In practice, the unconstrained optimum from SQ3 is unlikely to be realized.

Real-world decision-making rarely achieves perfect system-optimal outcomes. Exploring fixed offshore storage shares therefore helps reveal the system-level impacts and trade-offs that arise under different conditions. Two or three system contexts depending on SQ1 (for instance, a reference case, a tight-grid case, and a high CO₂-price case) are defined, and for each context a share ladder (e.g., 0/25/50/75/100% offshore) is simulated. Each constrained instance is compared with the unconstrained optimum for the same situation using multiple KPI's.

3.1.5 SQ5 - Synthesis

Finally, outputs from SQ3 and SQ4 are synthesized to characterize when offshore or onshore storage is favorable and what outcomes follow. The synthesis combines configuration trade-offs with condition maps, aggregating implications for KPI's. It draws on the KPI set compiled in SQ1 and will tie the findings back to the main question. Together, these findings will outline the role and implications of storage (siting) in shaping a future Dutch electricity system with large-scale offshore wind integration.

For an overview of the research approach, a research flow diagram is presented in Figure 1 below.

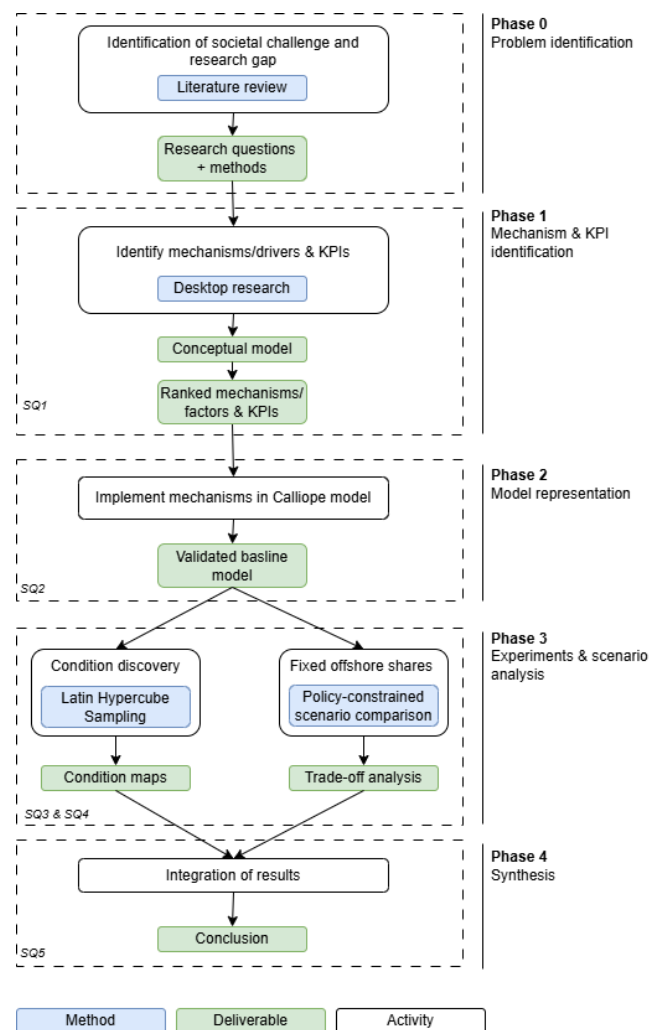


Figure 1 – Research Flow Diagram

4. Planning

In this Chapter, the planning for the research is shown using a Gantt chart in Figure 2. It is developed to ensure the thesis will be finalized in 21 working weeks.

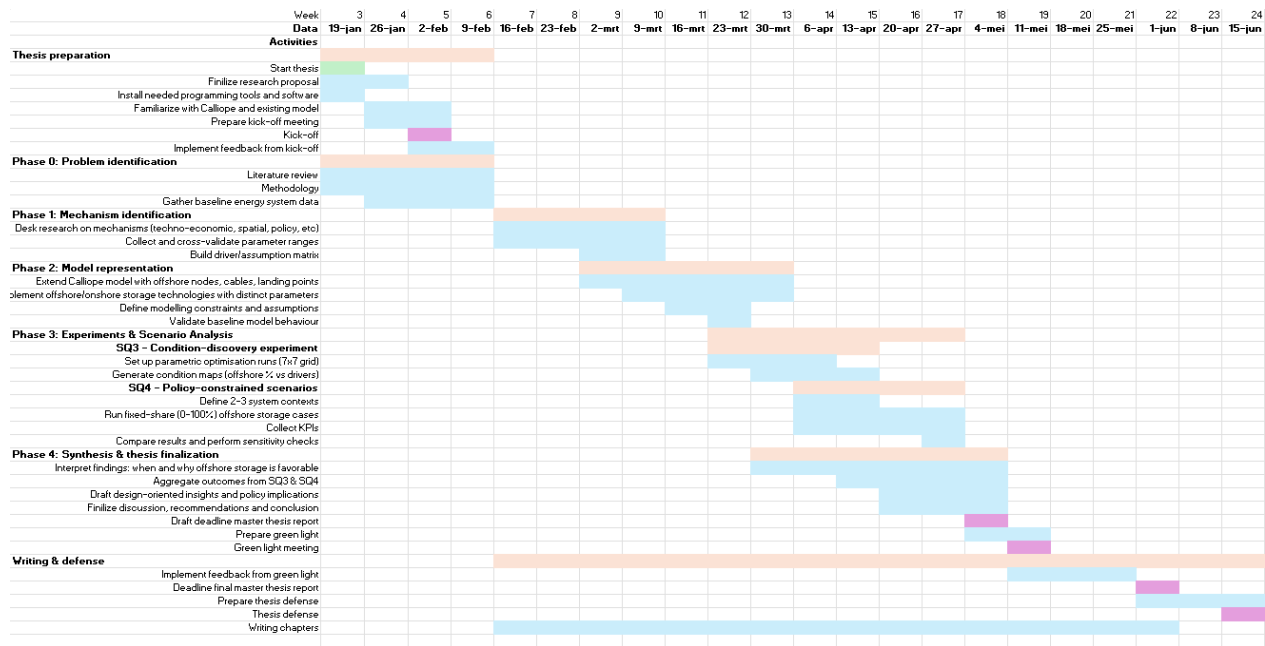


Figure 2 – Planning master thesis

Use of AI tools

During the writing of this report, ChatGPT was used for summarizing academic articles to work efficiently. The text was always reviewed, and the quoted articles were again scanned for a deeper understanding of its statements. Additionally, AI tools were used to improve the overall cohesion of the report. The text in the report is always written by the author, as ChatGPT was solely used to work efficiently. The author takes full responsibility for the content of this report.

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Appendix

This literature review used the simpler form of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 method. This systematic review method ensures that the selection of papers is based on trustworthiness and applicability in the research area, while results are as transparent and complete as possible, allowing others to retrieve the same results (Page et al., 2021). This method, consisting of preparing a search string and applying inclusion and exclusion criteria, limits the pool of papers in a structured way. This is shown visually in Figure A.1.

The Scopus database was used to find relevant articles. Before these results were screened for eligibility, non-English results were eliminated from the list. Screening is the second step of the PRISMA method. To reduce the number of articles, various exclusion criteria were applied to the remaining papers. After the exclusion criteria, possibly relevant papers are selected based on their title and abstract. The remaining papers are analyzed in more detail to come to a final selection. An overview of this final selection is provided in Table 1.

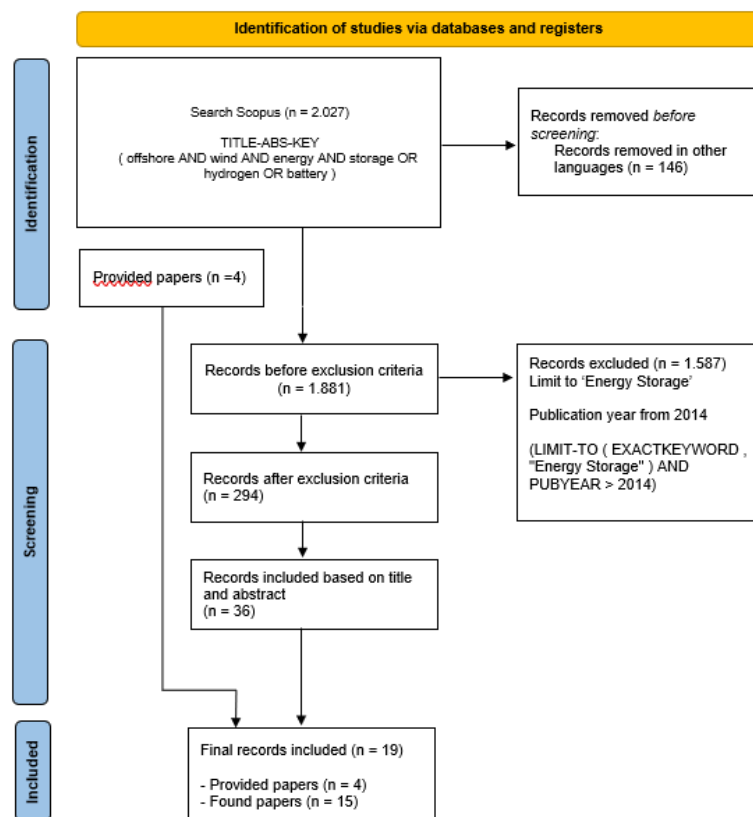


Figure A.1 – Flowchart (PRISMA)